V2G2T Program — Data Inventory

Deduplicated catalogue of all acquired datasets across the variant→gene→trait pipeline

Compiled 2026-07-12 | genome build hg38 throughout | single source of truth: **gs://claude-hackathon**

Executive summary

This inventory catalogues every dataset acquired for the V2G2T (variant→gene→trait) program, organised into six functional tiers and **deduplicated so that no record is counted twice**. The catalogue distinguishes canonical datasets from their derived views, byte-identical working copies, and dual-role uses (a full audit is in the deduplication ledger, p.3).

The **credible-set substrate** — the core V2G2T input — comprises **10,072,543 credible-set rows** (5,080,965 distinct variants; 48,411 at PIP>0.9), all on hg38, unified from four biobanks (AoU, UKB-WGS, FinnGen R13, BBJ) plus UKB NMR metabolomics. By layer: 7,837,680 molecular-QTL, 2,191,559 GWAS, 43,304 metabolite-QTL. This single combined table already contains all per-source files; the per-source parquets are slices of it, **not additional data**.

Around this substrate sit: molecular-QTL sets for colocalization (Open Targets in-sample SuSiE, UKB-PPP pQTL, non-European QTL), the DBNascent signed E2G training corpus and its ENCODE-rE2G comparator, two independent validation-truth sets (the 8-source V2G benchmark and the ENCODE-rE2G CRISPRi gold standard), the convergent gold V2G2T loci (Filters 1 & 2), and two gene→trait burden resources held for validation (AoU 392k, FinnGen R13 LoF).

Table 1 — Deduplicated data inventory (17 canonical datasets, 6 tiers)

Each row is one canonical dataset. Tier colour matches the scale figure on p.4. Where N is shown as “A / B” it denotes positives / negatives (validation sets).

Credible sets (V2G2T substrate)	**AoU MultiSuSiE**	GWAS	AFR+LAT+EUR	hg38	276,767	14 quant traits; WGS; multi-ancestry; 1,158 PIP>0.9	substrate
Credible sets (V2G2T substrate)	**UKB-WGS**	GWAS	EUR	hg38	471,513	34 traits; genuine WGS panel; 10,993 PIP>0.9	substrate
Credible sets (V2G2T substrate)	**UKB (TOPMed+UK10K imputation)**	GWAS	EUR	hg38	900,674	same 34 traits, imputation panels; provenance-tagged	substrate (imputation)
Credible sets (V2G2T substrate)	**FinnGen R13**	GWAS	FIN	hg38	542,605	1,424 endpoints; carries r2-to-lead + gene; 3,257 PIP>0.9	substrate
Credible sets (V2G2T substrate)	**BBJ / JCTF molQTL**	molQTL	EAS	hg38	7,837,680	cis-eQTL 19,026 genes + cis-pQTL 2,522 prot; in-sample; 5,725 PIP>0.9	substrate
Credible sets (V2G2T substrate)	**UKB metabolomics**	metaboliteQTL	multi-UKB	hg38	43,322	249 NMR traits; SuSiE; carries r2; 8,694 PIP>0.9	substrate
Molecular QTL (colocalization)	**OT molecular QTL (in-sample SuSiE)**	V2G/QTL	mostly EUR	hg38	2,012,086	eqtl 1.35M / tuqtl 385k / sqtl 224k / sceqtl 53k / pqtl 1.6k	QTL layer
Molecular QTL (colocalization)	**UKB-PPP pQTL (ST16, lifted)**	V2G/QTL	EUR	hg38	29,420	2,414 proteins; 10,750 cis / 18,670 trans; supersedes OT's 1,581	pQTL rescue
Molecular QTL (colocalization)	**Non-European QTL (7 OT projects)**	V2G/QTL	African/EAS/Admixed	hg38	158,208	GEUVADIS/Quach/Nedelec/Nathan/Perez/Randolph/iPSCORE	ancestry generalization
E2G training / comparator	**DBNascent signed pairs**	E2G training	multi-tissue	hg38	6,700,460	82.3% activating; 11 tissues; +per-tissue/negatives/3D views	signed E2G label
E2G training / comparator	**OT ENCODE-rE2G (distal)**	E2G comparator	—	hg38	10,370,321	18,575 genes / 322 biosamples; CRISPRi-trained; score non-selective	comparator / feature
Validation truth	**V2G benchmark v1.0 (8 sources)**	validation	multi	hg38	2,164 / 13,613	kanai/cortex-MPRA/schnitzler/tardaguila/ghatan/FAIR/STING/autoimmune; 4 assay classes	held-out truth
Validation truth	**ENCODE-rE2G CRISPRi (Gschwind 2023)**	E2G truth	K562+	hg38	10,411 / 1,075	concordance-test truth (666 matched) AND comparator source — ONE dataset	label-validity gate
Gold V2G2T loci	**V2G2T Filter 1 (LD non-coding + pLoF)**	gold V2G2T	FIN (in-sample)	hg38	86 (≤250kb)	53 distinct v→g; ABO/CHEK2/LZTR1/RAD52/ICAM1	assay-independent validation
Gold V2G2T loci	**V2G2T Filter 2 (burden+GWAS+coloc)**	gold V2G2T	multi	hg38	166	133 distinct v→g / 84 genes; GRN/APOB/LDLR/TP53/CFH	assay-independent validation
G2T burden (companion)	**AoU 392k All-by-All burden**	G2T	multi (AoU)	hg38	1,580	235 genes / 327 phenotypes; SD3+SD5+SD7	G2T validation
G2T burden (companion)	**FinnGen R13 LoF burden**	G2T	FIN	hg38	13,968	3,984 genes; 304 genome-wide significant	G2T validation

Table 2 — Deduplication ledger (redundancies resolved, no double-counting)

The specific redundancies identified across the GWAS / CRISPR / benchmark data and the rule applied to each, so that headline counts in this report are auditable.

Combined vs per-source credible sets	multibiobank_credible_sets_combined.parquet (10,072,543 rows) = concatenation of the 6 per-source files	Count credible sets ONCE via the combined table; per-source files are slices, not additional data	10,072,543 combined = sum of parts; do not add
Byte-identical benchmark duplicates	negatives.parquet ≡ v2g_benchmark_negatives.parquet; positives_all10.parquet ≡ v2g_benchmark_positives.parquet (MD5-identical)	Keep v2g_benchmark_* as canonical; the bare names are earlier working copies	13,613 neg / 2,164 pos counted once
Positives working versions	positives_6sources / positives_canonical / positives_all10 / v2g_benchmark_positives_enriched are stages of one build	Canonical = v2g_benchmark_positives(_enriched); 2,164 positives total	2,164 (not 2,164×4)
CRISPRi ENCODE-rE2G double role	Used as (a) concordance-test truth (666 matched pairs) and (b) inventory comparator	ONE dataset (10,411 tested / 1,075 positive); two USES, not two datasets	1,075 positive pairs counted once
DBNascent derived views	signed pairs (6.7M) / per-tissue obs (12.7M) / negatives (9.98M) / 3D-stringent (2.73M) are views of ONE source	One source, four training views; do not sum the four Ns	6.7M unique signed pairs is the base unit
OT ENCODE-rE2G all vs distal	48.8M all links vs 10.37M distal — distal is the analysis subset	Report distal (10.37M) as the used set; 48.8M is the raw superset	10,370,321 distal used
Benchmark sources vs training data	V2G benchmark 8 sources (kanai/MPRA/etc.) are DISTINCT from DBNascent training + OT QTL	No overlap — benchmark is held-out; kept separate by design	independent
Metabolite-GWAS proxy superseded	Old OT metabolite-keyword proxy (144,418 CS) replaced by real UKB metabolomics SuSiE CS (43,322)	Use UKB metabolomics fine-mapped set; drop the keyword proxy from the substrate	proxy retired
Kanai cross-biobank fine-mapping	Dropped per user (noisy 94-trait imputation fine-mapping) — but kanai_multiome_2025 in the benchmark is a DIFFERENT curated set	Excluded from substrate; benchmark's kanai_multiome is a separate validated-triplet source	excluded from substrate

Figure 1 — Canonical dataset scale by tier

Record counts per canonical dataset (log scale). Legend shown as a separate key to the right so the plot keeps its full data area.

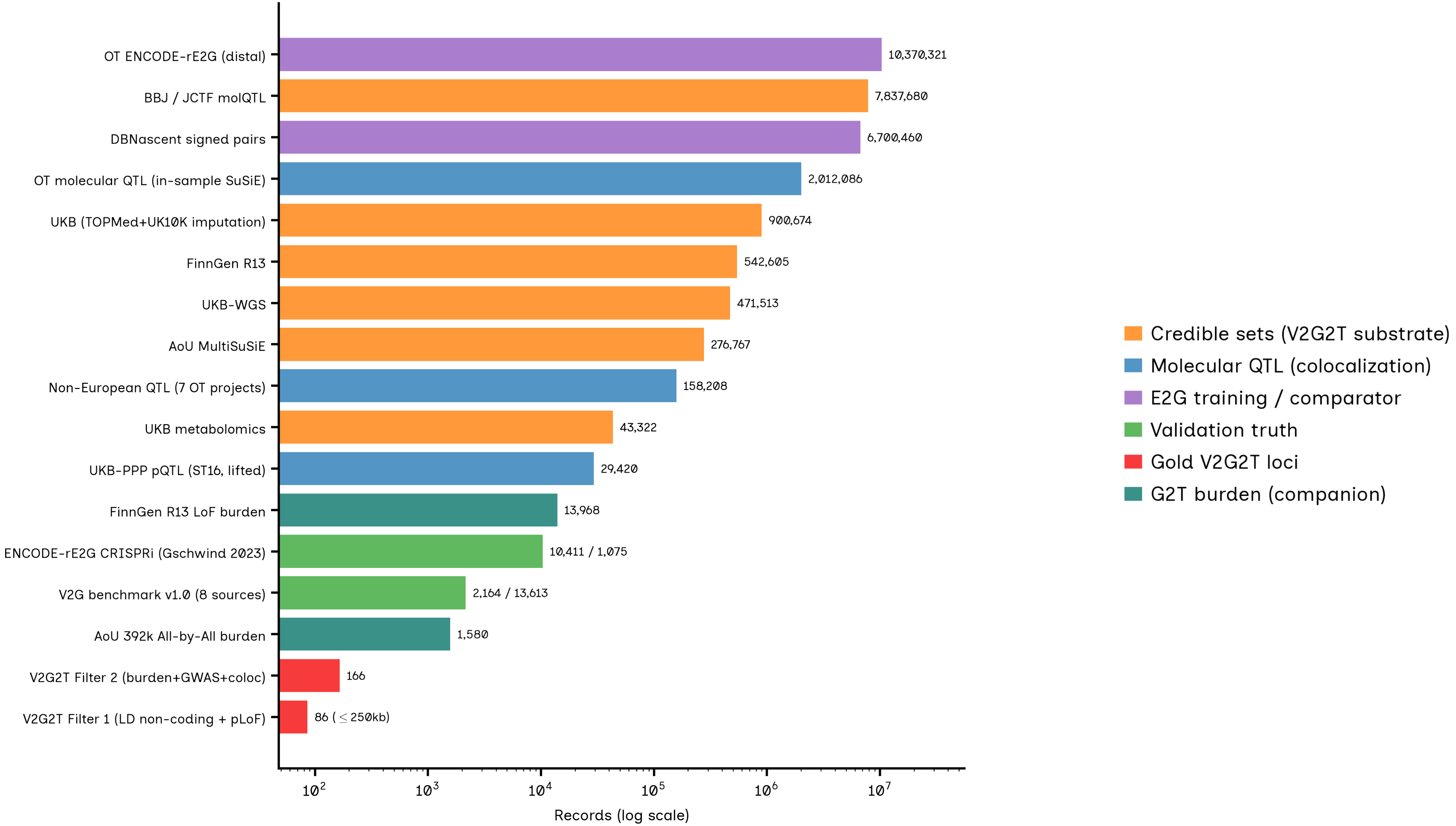
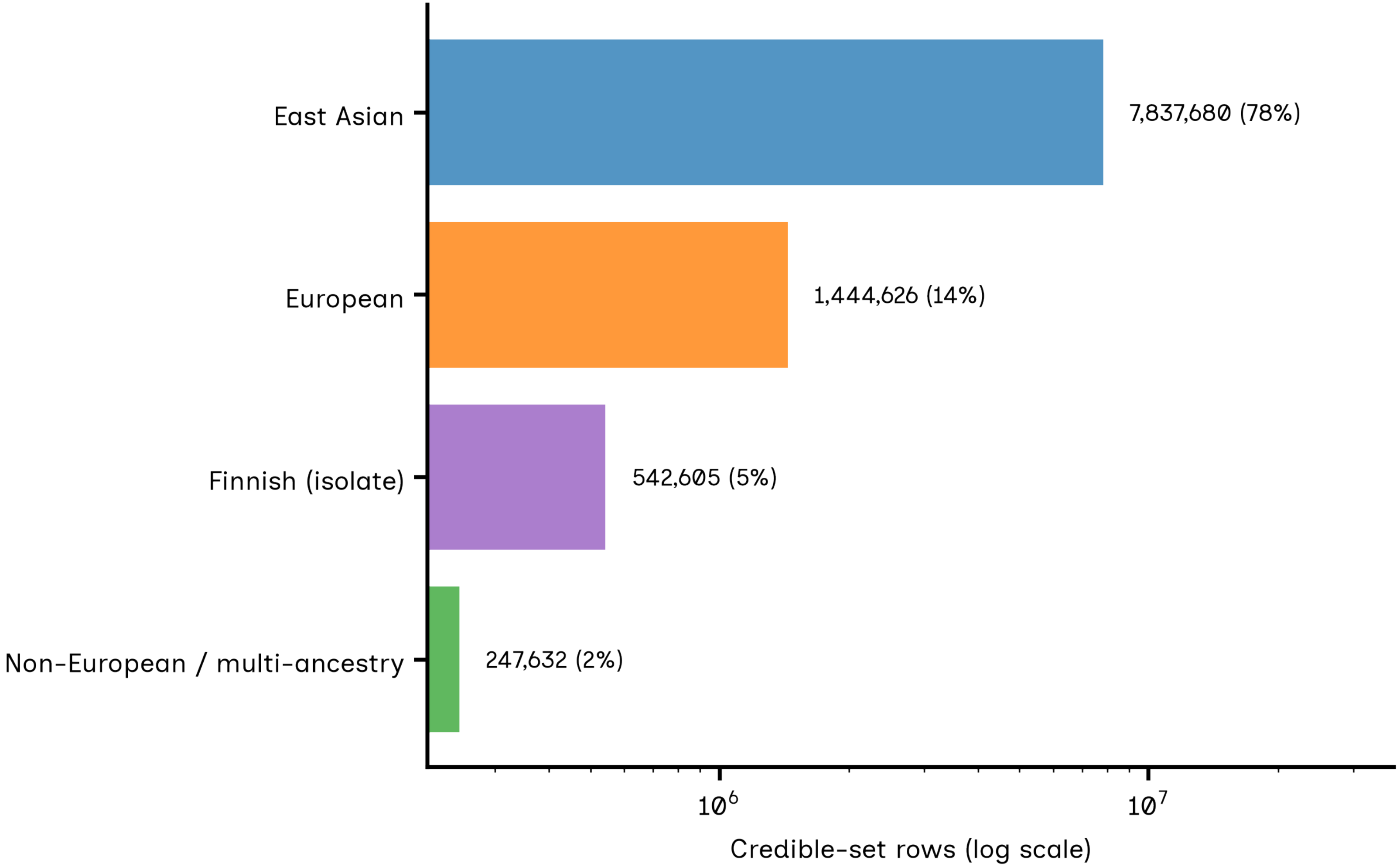


Figure 2 — Ancestry representation across the credible-set substrate

Credible-set rows by broad ancestry (log scale). The East-Asian share is dominated by BBJ molecular-QTL members (7.84M cis-e/pQTL); for GWAS-only diversity, the meaningful gains are AoU multi-ancestry and FinnGen (isolate).



Provenance, staging & exclusions

Genome build. All datasets are hg38. UKB-PPP pQTL (Sun 2023) was lifted hg19→hg38 (100% mapped; 69% cis / 85% trans exact-match validated against the OT hg38 universe).

Assay basis. Each credible set is tagged wgs vs imputation. Genuine WGS: AoU (multi-ancestry) and UKB-WGS (471,513 rows). Imputation: FinnGen (SISu), BBJ (JCTF), and the UKB TOPMed/UK10K provenance slices. This tag is first-class so imputation-based fine-mapping is never silently mixed with WGS.

In-sample LD requirement. Every credible set derives from in-sample-LD SuSiE/MultiSuSiE fine-mapping. Sets built on reference-panel LD were excluded on principle.

Staging. All parquets live at **gs://claude-hackathon/multibiobank/20260711/** (combined + persource/ + qc/) and as project artifacts. Acquisition/harmonization scripts and QC reports are versioned in the GitHub repo (scripts/multibiobank/, docs/qc/).

Explicit exclusions. (1) **Kanai 94-trait UKB fine-mapping** — dropped as legacy imputation-panel fine-mapping the user found noisy; note the benchmark's *kanai_multiome_2025* is a separate curated validated-triplet source, not the same data. (2) **Multi-cohort meta-pQTL (Cell 2026)** — excluded: 38-study meta-analysis on reference LD, fails the in-sample gate. (3) **OT metabolite-keyword proxy** (144,418 CS) — retired in favour of the real UKB NMR SuSiE credible sets (43,322). (4) **FinnGen native pairwise LD matrix** — the public release ships only the SISu v4.2 reference panel (n=3,775), so per-CS r^2-to-lead was captured but the full pairwise matrix was not used.

Headline counts are deduplicated

Credible-set substrate = 10,072,543 rows counted **once** via the combined table (per-source files are slices). CRISPRi ENCODE-rE2G is one dataset used in two roles (concordance truth + comparator). DBNascent is one source with four training views (6.7M unique signed pairs is the base unit). Byte-identical working copies (negatives.parquet, positives_all10.parquet) are collapsed to their v2g_benchmark_* canonicals.